Daily Algebra Drill #2 — Answers & Investigations

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Vieta's formulas for cubics · Nested surds · Telescoping products · Brahmagupta identity · Factor theorem · Cyclic sums under constraints.

Section A Rapid Recognition (≤15 s each)

Factorise, simplify, or evaluate instantly. No expansion.

1. Factorise completely: $x^3 - 27$.

Answer: $(x-3)(x^2+3x+9)$

Method: Sum/difference of cubes: $a^3 - b^3 = (a-b)(a^2+ab+b^2)$ with $a = x$, $b = 3$. The quadratic factor is irreducible over $\mathbb{R}$.

> **Investigate further:** Show that $a^3 - b^3 = (a-b)(a^2+ab+b^2)$ by expanding the right-hand side. Then generalise: for which values of n does $(a-b)$ divide $a^n - b^n$? (*Answer: all positive integers n.*) Prove it using the factor theorem.

2. Without expanding, evaluate $(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})$.

Answer: 2

Method: Conjugate product: $(a+b)(a-b) = a^2 - b^2 = 5 - 3 = 2$. This is the core move for rationalising surds.

> **Investigate further:** Evaluate $(\sqrt{n+1}+\sqrt{n})(\sqrt{n+1}-\sqrt{n})$ for any positive integer n. Use this to write $\frac{1}{\sqrt{n+1}+\sqrt{n}}$ in a simpler form, then telescope $\sum_{n=1}^{99} \dfrac{1}{\sqrt{n+1}+\sqrt{n}}$.

3. Without a calculator, evaluate $999^2 - 1$.

Answer: 998 000

Method: $999^2 - 1^2 = (999+1)(999-1) = 1000 \times 998 = 998\,000$. Treat every "near-square minus 1" as a difference of two squares.

> **Investigate further:** Prove that every product of two integers whose average is an integer can be written as a perfect square minus a perfect square. (Hint: if $p + q = 2m$, write $pq = m^2 - \left(\frac{p-q}{2}\right)^2$.) This is the basis of Fermat's factorisation method.

4. Given $a - b = 4$ and $ab = 5$, find $a^2 + b^2$.

Answer: 26

Method: $(a-b)^2 = a^2 - 2ab + b^2 = 16$, so $a^2 + b^2 = 16 + 2(5) = 26$. Never solve for a and b individually.

> **Investigate further:** Form the quadratic with roots a and b: you know $a - b = 4$ and $ab = 5$, so $a + b = \pm\sqrt{(a-b)^2 + 4ab} = \pm\sqrt{36} = \pm 6$. Find $a^3 - b^3$ and $a^4 + b^4$ without finding a or b.

5. Expand and simplify: $(x+y)^3 - x^3 - y^3$.

Answer: $3xy(x+y)$

Method: Expand $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ then cancel x^3 and y^3. Factor out $3xy$. Memorise: $(x+y)^3 - x^3 - y^3 = 3xy(x+y)$.

> **Investigate further:** This is the key ingredient in the identity $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$. Verify this identity by expanding, then use it to prove that if $a+b+c=0$ then $a^3+b^3+c^3 = 3abc$.

6. Factorise completely: $3x^2 - 75$.

Answer: $3(x-5)(x+5)$

Method: Take out the common factor of 3 first, giving $3(x^2-25)$, then apply difference of two squares. Always extract scalars before factorising.

> **Investigate further:** Generalise: find all positive integers k for which $kx^2 - k \cdot n^2$ factors into three factors over the integers. When does $ax^2 + bx + c$ have a common integer factor across all three coefficients?

7. Rationalise the denominator: $\dfrac{1}{\sqrt{3}+\sqrt{2}}$.

Answer: $\sqrt{3} - \sqrt{2}$

Method: Multiply top and bottom by the conjugate $(\sqrt{3}-\sqrt{2})$. Denominator becomes $3-2=1$. The result is exact and requires no further work.

> **Investigate further:** Show that $\dfrac{1}{\sqrt{n}+\sqrt{n-1}} = \sqrt{n} - \sqrt{n-1}$. Hence find a closed form for $\displaystyle\sum_{n=1}^{N} \frac{1}{\sqrt{n}+\sqrt{n-1}}$.

8. Factorise: $x^2(x-1) - (x-1)$.

Answer: $(x-1)(x-1)(x+1) = (x-1)^2(x+1)$

Method: $(x-1)$ is a common factor: $(x-1)(x^2-1) = (x-1)(x-1)(x+1)$. Always look for a common binomial factor before anything else.

> **Investigate further:** Use the factor theorem: verify that $x=1$ is a double root of $x^3 - x^2 - x + 1$ (which equals $x^2(x-1)-(x-1)$) by showing $f(1)=0$ and $f'(1)=0$. What does a double root mean geometrically on the graph of f?

9. The roots of $x^2 - 5x + 3 = 0$ are p and q. Find p^2+q^2.

Answer: 19

Method: By Vieta: $p+q=5$, $pq=3$. Then $p^2+q^2 = (p+q)^2 - 2pq = 25 - 6 = 19$. This is Vieta + Newton identity: essential toolkit.

> **Investigate further: Vieta's formulas for a cubic.** If α, β, γ are roots of $x^3 + px^2 + qx + r = 0$, then $\alpha+\beta+\gamma = -p$, $\alpha\beta + \beta\gamma + \gamma\alpha = q$, $\alpha\beta\gamma = -r$. Use these to find $\alpha^2+\beta^2+\gamma^2$ in terms of p

and q. (Answer: $p^2 - 2q$.) Verify with the cubic $x^3 - 6x^2 + 11x - 6 = 0$.

10. Simplify: $\dfrac{2^{n+3} - 2^n}{2^n}$.

Answer: 7

Method: Factor 2^n from the numerator: $\dfrac{2^n(2^3-1)}{2^n} = 8 - 1 = 7$. Powers of 2 always simplify by factoring out the smallest power.

Investigate further: Generalise: simplify $\dfrac{k^{n+m} - k^n}{k^n}$ for any base k and exponent m. Then investigate $\dfrac{2^{2n+1} + 2^{2n-1}}{2^{2n}}$: can you spot the pattern for alternating-parity exponents?

Section B Manipulation Drills

($\leq$50 s each)

Fractions, rearranging, hidden factorisations. Elegance over brute force.

1. Simplify: $\dfrac{x}{x^2-1} + \dfrac{1}{x-1}$.

Answer: $\dfrac{2x+1}{(x-1)(x+1)}$

Method: Common denominator is $(x-1)(x+1)$. Numerator: $x + (x+1) = 2x + 1$. Do not expand the denominator.

Investigate further: Now add a third term: simplify $\dfrac{x}{x^2-1} + \dfrac{1}{x-1} - \dfrac{1}{x+1}$. Then investigate: for which x is the original expression negative?

2. Factorise: $x^4 - x^2 - 12$.

Answer: $(x^2-4)(x^2+3) = (x-2)(x+2)(x^2+3)$

Method: Let $u = x^2$: $(u-4)(u+3) = 0$. Only $u = 4$ gives real solutions. The factor (x^2+3) is irreducible over $\mathbb{R}$.

Investigate further: Find all real solutions of $x^4 - x^2 - 12 = 0$. Then modify the constant: for what values of c does $x^4 - x^2 + c = 0$ have (i) four real roots, (ii) two real roots, (iii) no real roots? (*Substitute $u = x^2$ and analyse the discriminant.*)

3. Make r the subject of $S = \dfrac{a}{1-r}$.

Answer: $r = 1 - \dfrac{a}{S} = \dfrac{S-a}{S}$

Method: $S(1-r) = a \Rightarrow 1 - r = a/S \Rightarrow r = 1 - a/S$. This is the geometric series sum formula — essential for TMUA Paper 1.

Investigate further: Geometric series connection. The formula $S = \frac{a}{1-r}$ holds for $|r| < 1$. Starting from $S_n = a\frac{1-r^n}{1-r}$, make r the subject in terms of S_n, a, n. Why is there no closed-form solution in general? (This links to polynomial root-finding and the Abel–Ruffini theorem.)

4. Simplify: $\left(1+\dfrac{1}{x}\right)\left(1-\dfrac{1}{x+1}\right)$.

Answer: $\dfrac{x+1}{x}\cdot\dfrac{x}{x+1}=1$

Method: Write each factor as a single fraction: $\frac{x+1}{x}\cdot\frac{x}{x+1}$. The product telescopes to 1. Recognise the cancellation *before* multiplying out.

Investigate further: Generalise: show that $\prod_{k=1}^{n}\left(1+\dfrac{1}{k}\right)\left(1-\dfrac{1}{k+1}\right)$ telescopes. Then evaluate $\prod_{k=2}^{n}\left(1-\dfrac{1}{k^2}\right)$ by factoring each term as $\frac{(k-1)(k+1)}{k^2}$ and telescoping.

5. Given $x+y=3$ and $xy=1$, find x^3+y^3.

Answer: 18

Method: $x^3+y^3=(x+y)(x^2-xy+y^2)=(x+y)\big[(x+y)^2-3xy\big]=3(9-3)=18$. Build up from symmetric functions — never solve for x and y.

Investigate further: Find x^4+y^4 and x^5+y^5 using the same data ($x+y=3$, $xy=1$). (*Key: Newton's identity* $p_n=(x+y)p_{n-1}-xy\cdot p_{n-2}$ gives a recurrence $p_n=3p_{n-1}-p_{n-2}$.) Compute $p_1,p_2,\ldots,p_6$ and spot any patterns modulo small integers.

6. Simplify: $\dfrac{(a+b)^2-(a-b)^2}{4ab}$.

Answer: 1

Method: $(a+b)^2-(a-b)^2=4ab$ by the identity $(P+Q)^2-(P-Q)^2=4PQ$. The expression equals $4ab/4ab=1$.

Investigate further: The identity $(a+b)^2-(a-b)^2=4ab$ can be rewritten as $\left(\dfrac{a+b}{2}\right)^2-\left(\dfrac{a-b}{2}\right)^2=ab$, which says $\text{AM}^2-\text{HM-related term}^2=ab$. Use this to prove the AM–GM inequality $\frac{a+b}{2}\geq\sqrt{ab}$ for $a,b>0$ by observing that a square is always ≥ 0.

7. Factorise: x^3+3x^2-x-3.

Answer: $(x-1)(x+1)(x+3)$

Method: Group: $(x^3+3x^2)-(x+3)=x^2(x+3)-(x+3)=(x+3)(x^2-1)=(x+3)(x-1)(x+1)$. Grouping is faster than the factor theorem here.

Investigate further: Use the factor theorem: check $x=1,-1,-3$ in turn. Then compare the grouping method versus synthetic division — which is faster? Generalise: $x^3+ax^2-x-a=(x^2-1)(x+a)$ for any a. For which cubics does grouping always work?

8. Solve: $\dfrac{x-1}{x+2}=\dfrac{x+3}{x+8}$.

Answer: $x=7$

Method: Cross-multiply: $(x-1)(x+8) = (x+3)(x+2)$. Expand both sides: $x^2+7x-8 = x^2+5x+6$. The x^2 terms cancel: $2x = 14$, so $x = 7$. Check: LHS $= \frac{6}{9} = \frac{2}{3}$, RHS $= \frac{10}{15} = \frac{2}{3}$.

> **Investigate further:** Notice the x^2 terms always cancel when both sides are linear-over-linear with the same degree numerator and denominator. Investigate: for which values of a, b, c, d does $\dfrac{x+a}{x+b} = \dfrac{x+c}{x+d}$ have no solutions? Exactly one? Infinitely many?

9. Simplify: $\dfrac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}} + \dfrac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$.

> **Answer:** $\dfrac{2(a+b)}{a-b}$

Method: Let $P = \sqrt{a}+\sqrt{b}$, $Q = \sqrt{a}-\sqrt{b}$. The expression is $\frac{Q}{P} + \frac{P}{Q} = \frac{P^2+Q^2}{PQ}$. $P^2+Q^2 = 2(a+b)$, $PQ = a-b$.

> **Investigate further:** Compare this with the structure $t + \frac{1}{t}$ where $t = \frac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}}$. Note $t \in (-1, 1)$ for $a, b > 0$. Show that $t + 1/t \geq 2$ is false when $|t| < 1$ — this is why AM–GM requires *positive* quantities. What is the minimum of $\frac{2(a+b)}{a-b}$ subject to $a > b > 0$?

10. The cubic $x^3 - 6x^2 + 11x - 6 = 0$ has roots α, β, γ. Without solving it, find $\alpha^2+\beta^2+\gamma^2$ and $\alpha\beta\gamma$.

> **Answer:** $\alpha^2+\beta^2+\gamma^2 = 14; \quad \alpha\beta\gamma = 6$

Method: By Vieta's formulas: $\alpha+\beta+\gamma = 6$, $\alpha\beta+\beta\gamma+\gamma\alpha = 11$, $\alpha\beta\gamma = 6$. Then $\alpha^2+\beta^2+\gamma^2 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha) = 36 - 22 = 14$.

> **Investigate further: Newton's power sums for a cubic.** Define $p_k = \alpha^k+\beta^k+\gamma^k$. You found $p_1 = 6$ and $p_2 = 14$. Use Newton's identity: $p_k = (\alpha+\beta+\gamma)p_{k-1} - (\alpha\beta+\beta\gamma+\gamma\alpha)p_{k-2} + \alpha\beta\gamma \cdot p_{k-3}$ to find p_3 without solving the cubic. Verify by checking $\alpha = 1, \beta = 2, \gamma = 3$.

Section C Substitution & Structure (≤90 s each)

Find structure before computing. Substitution is the key, not expansion.

1. Solve: $x^4 - 13x^2 + 36 = 0$.

> **Answer:** $x = \pm 2,\ \pm 3$

Method: Let $u = x^2$: $(u-4)(u-9) = 0$, giving $u = 4$ or $u = 9$. Both positive, so four real solutions $x = \pm 2, \pm 3$.

> **Investigate further:** Now solve $x^4 - 13x^2 + 36 = k$ for $k = 10$, $k = -1$. For which values of k does the equation have 4, 2, or 0 real solutions? Sketch $y = x^4 - 13x^2 + 36$ and identify the local minima and maxima algebraically (without calculus) by completing the square in $u = x^2$.

2. Solve: $(x+1)(x+2)(x+3)(x+4) = 24$.

> **Answer:** $x = 0$ and $x = -5$

Method: Pair the outer and inner terms: $(x+1)(x+4) = x^2+5x+4$ and $(x+2)(x+3) = x^2+5x+6$. Let $u = x^2+5x+5$: $(u-1)(u+1) = 24 \Rightarrow u^2 = 25 \Rightarrow u = \pm 5$. $u = 5$: $x^2+5x = 0 \Rightarrow x(x+5) = 0$. $u = -5$: $x^2+5x+10 = 0$, discriminant < 0, no real solutions.

> **Investigate further:** This "pair the ends" trick works whenever the four roots are in arithmetic progression. Generalise: solve $(x+a)(x+b)(x+c)(x+d) = k$ where $a+d = b+c$. Then try: $(x)(x+2)(x+4)(x+6) = 9$.

3. Simplify: $\sqrt{6+2\sqrt{5}}$.

Answer: $\sqrt{5}+1$

Method: Write $6+2\sqrt{5} = 5+2\sqrt{5}+1 = (\sqrt{5}+1)^2$. Then $\sqrt{(\sqrt{5}+1)^2} = \sqrt{5}+1$ (positive root).

> **Investigate further: Denesting nested surds.** To simplify $\sqrt{a+b\sqrt{c}}$, guess the form $\sqrt{p}+\sqrt{q}$ and solve $p+q=a$, $4pq = b^2c$. Use this method to simplify $\sqrt{7+2\sqrt{10}}$ and $\sqrt{11-4\sqrt{7}}$. When does $\sqrt{a+b\sqrt{c}}$ denest to a rational combination of simple surds?

4. Given $a+b+c=1$ and $a^2+b^2+c^2=1$, find $ab+bc+ca$ and hence $a^3+b^3+c^3-3abc$.

Answer: $ab+bc+ca=0; \quad a^3+b^3+c^3-3abc=1$

Method: $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca) \Rightarrow 1 = 1+2(ab+bc+ca)$, so $ab+bc+ca=0$. Then apply the factorisation $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 1\cdot(1-0) = 1$. Note abc itself is *not* determined by the given data — only the combination $a^3+b^3+c^3-3abc$ is.

> **Investigate further:** Use Newton's identity $p_3 = e_1p_2 - e_2p_1 + 3e_3$ (where e_1, e_2, e_3 are the elementary symmetric polynomials) to find $a^3+b^3+c^3$ in terms of e_1, e_2, e_3 in general. Verify with $a=1, b=0, c=0$. Then investigate: if $a^n+b^n+c^n$ is an integer for $n=1,2,3$, must it be an integer for all n? (Ramanujan connection.)

5. Solve: $(x-1)(x-3)(x-5)(x-7)+15=0$.

Answer: $x = 2,\ 6,\ 4\pm\sqrt{6}$ (all four real)

Method: Pair: $(x-1)(x-7) = x^2-8x+7$ and $(x-3)(x-5) = x^2-8x+15$. $(u-4)(u+4) = u^2-16$, so $u^2-16+15 = u^2-1 = 0 \Rightarrow u=\pm1$. $u=1$: $x^2-8x+10=0 \Rightarrow x = 4\pm\sqrt{6}$. $u=-1$: $x^2-8x+12=0 \Rightarrow (x-2)(x-6)=0$.

> **Investigate further:** The "pair the ends" substitution $u = x^2-8x+\text{const}$ works because the four linear factors are symmetric about $x=4$ (the mean of $1,3,5,7$). Investigate: $(x-a)(x-b)(x-c)(x-d)+k=0$ where $a+d=b+c$. Show that the substitution $u = x^2-(a+d)x+\frac{(a+d)^2}{4}$ always reduces this to a quadratic in u.

6. If $x = 2+\sqrt{3}$, find $x^2+\dfrac{1}{x^2}$ without expanding x^2 directly.

Answer: 14

Method: Note $\frac{1}{x} = \frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$ (conjugate rationalisation). So $x+\frac{1}{x} = (2+\sqrt{3})+(2-\sqrt{3}) = 4$. Then $x^2+\frac{1}{x^2} = \left(x+\frac{1}{x}\right)^2-2 = 16-2 = 14$.

> **Investigate further:** Observe that $x = 2+\sqrt{3}$ and $\frac{1}{x} = 2-\sqrt{3}$ are conjugate surds. Show that for $x = a+\sqrt{b}$ (with a, b rational), $\frac{1}{x} = \frac{a-\sqrt{b}}{a^2-b}$. When does $x+\frac{1}{x}$ become an integer? Find all integers n such that $x+\frac{1}{x} = n$ for some $x = a+\sqrt{b}$ with $a, b\in\mathbb{Z}$.

7. Given $\dfrac{a}{b}+\dfrac{b}{a}=3$, find $\dfrac{a^2}{b^2}+\dfrac{b^2}{a^2}$.

Answer: 7

Method: Let $t=\frac{a}{b}+\frac{b}{a}=3$. Then $\frac{a^2}{b^2}+\frac{b^2}{a^2}=\left(\frac{a}{b}+\frac{b}{a}\right)^2-2=9-2=7$.

> **Investigate further:** Extend the chain: find $\dfrac{a^3}{b^3}+\dfrac{b^3}{a^3}$ using the identity $t^3=\left(\frac{a}{b}\right)^3+\left(\frac{b}{a}\right)^3+3\cdot\frac{a}{b}\cdot\frac{b}{a}\left(\frac{a}{b}+\frac{b}{a}\right)$. Generalise: if $s_n=\left(\frac{a}{b}\right)^n+\left(\frac{b}{a}\right)^n$, show that $s_n=3s_{n-1}-s_{n-2}$ and compute s_1,s_2,s_3,s_4,s_5.

8. If $x+y+z=0$, simplify $\dfrac{(x^2+y^2+z^2)^2}{x^4+y^4+z^4}$.

Answer: 2

Method: Let $s=x^2+y^2+z^2$. Since $x+y+z=0$: $(x+y+z)^2=0\Rightarrow x^2+y^2+z^2=-2(xy+yz+zx)$, so $xy+yz+zx=-s/2$. Then $(x^2+y^2+z^2)^2=x^4+y^4+z^4+2(x^2y^2+y^2z^2+z^2x^2)$. Also $(xy+yz+zx)^2=x^2y^2+y^2z^2+z^2x^2+2xyz(x+y+z)=x^2y^2+y^2z^2+z^2x^2$. So $s^2=x^4+y^4+z^4+2\cdot s^2/4$, giving $x^4+y^4+z^4=s^2/2$. The ratio is $s^2/(s^2/2)=2$.

> **Investigate further:** This result says that under the constraint $x+y+z=0$, the ratio of the square of the second power sum to the fourth power sum is always 2. Investigate the analogous ratio $\dfrac{(x^2+y^2+z^2)^3}{x^6+y^6+z^6}$ when $x+y+z=0$. Also: prove that $x^4+y^4+z^4=\frac{1}{2}(x^2+y^2+z^2)^2$ is equivalent to $(xy+yz+zx)^2=2xyz(x+y+z)$, which is automatic when the sum is zero.

Section D Challenge (TMUA / SMC difficulty)

Each has a solution under five lines. Find the insight, not the computation.

1. Evaluate:

$$\frac{1}{1\cdot2\cdot3}+\frac{1}{2\cdot3\cdot4}+\frac{1}{3\cdot4\cdot5}+\cdots+\frac{1}{8\cdot9\cdot10}.$$

Answer: $\dfrac{11}{45}$

Method: Partial fractions: $\dfrac{1}{n(n+1)(n+2)}=\dfrac{1}{2}\left(\dfrac{1}{n(n+1)}-\dfrac{1}{(n+1)(n+2)}\right)$. This is a telescoping sum! Sum from $n=1$ to 8: $\dfrac{1}{2}\left(\dfrac{1}{1\cdot2}-\dfrac{1}{9\cdot10}\right)=\dfrac{1}{2}\left(\dfrac{1}{2}-\dfrac{1}{90}\right)=\dfrac{1}{2}\cdot\dfrac{44}{90}=\dfrac{22}{90}=\dfrac{11}{45}$.

> **Investigate further: Higher-order telescoping.** Generalise the partial fraction: $\dfrac{1}{n(n+1)\cdots(n+k)}=\dfrac{1}{k}\left(\dfrac{1}{n(n+1)\cdots(n+k-1)}-\dfrac{1}{(n+1)\cdots(n+k)}\right)$. Use this to find a closed form for $\displaystyle\sum_{n=1}^{N}\frac{1}{n(n+1)(n+2)(n+3)}$ and verify for $N=2$.

2. The roots of $x^2+px+q=0$ are r and s. Find, in terms of p and q, the monic quadratic whose roots are r^2+s and s^2+r.

Answer: $x^2-(p^2-2q-p)\,x+(q^2-p^3+3pq+q)=0$

Method: By Vieta on the original: $r+s=-p$, $rs=q$. New sum: $(r^2+s)+(s^2+r)=(r^2+s^2)+(r+s)=$

$[(r+s)^2-2rs]+(r+s) = (p^2-2q)+(-p) = p^2-2q-p$. New product: $(r^2+s)(s^2+r) = r^2s^2+r^3+s^3+rs = q^2+(r+s)^3-3rs(r+s)+q = q^2+(-p)^3-3q(-p)+q = -p^3+3pq+q^2+q$. Required quadratic: $x^2-(p^2-2q-p)x+(-p^3+3pq+q^2+q) = 0$.

> **Investigate further:** This is a classic example of constructing a new polynomial from transformed roots — a technique called *Tschirnhaus transformation.* Try the simpler version: find the quadratic with roots r^2 and s^2. Then find the quartic with roots r^2, s^2, r^3, s^3 (this is much harder — think about what Vieta gives you for degree 4).

3. The equation $\dfrac{x}{y+z} = \dfrac{y}{x+z} = \dfrac{z}{x+y} = k$ holds for real x, y, z with $x+y+z \neq 0$. Find all possible values of k.

Answer: Under $x+y+z \neq 0$: $k = \frac{1}{2}$ only.

Method: Set each ratio equal to k: $x = k(y+z)$, $y = k(x+z)$, $z = k(x+y)$. Adding gives $x+y+z = 2k(x+y+z)$. Since $x+y+z \neq 0$, divide through to get $k = \frac{1}{2}$.

> **Investigate further:** Investigate the case $x+y+z = 0$ separately: show that each ratio equals -1. (When $x+y+z = 0$, we have $y+z = -x$, so $\frac{x}{y+z} = \frac{x}{-x} = -1$.) Hence in general, without any constraint, the possible values of k are $\frac{1}{2}$ and -1. Find $(x : y : z)$ for each case.

4. Prove the **Brahmagupta–Fibonacci identity**:

$$(a^2+b^2)(c^2+d^2) = (ac+bd)^2 + (ad-bc)^2.$$

Hence express 85×65 as a sum of two perfect squares in two different ways.

Answer: $85 \times 65 = 5525 = 74^2 + 7^2 = 71^2 + 22^2$.

Method: *Proof:* Expand the right-hand side: $(ac+bd)^2 + (ad-bc)^2 = a^2c^2 + 2abcd + b^2d^2 + a^2d^2 - 2abcd + b^2c^2 = a^2(c^2+d^2) + b^2(c^2+d^2) = (a^2+b^2)(c^2+d^2)$. □

Representations: Use $85 = 2^2+9^2$, and $65 = 1^2+8^2$ and $65 = 4^2+7^2$. $(2^2+9^2)(1^2+8^2) = (2\cdot1+9\cdot8)^2 + (2\cdot8-9\cdot1)^2 = 74^2+7^2 = 5476+49 = 5525$. ✓ $(2^2+9^2)(4^2+7^2) = (8+63)^2+(14-36)^2 = 71^2+22^2 = 5041+484 = 5525$. ✓

> **Investigate further:** The Brahmagupta–Fibonacci identity has a beautiful complex-number proof: $|z_1|^2|z_2|^2 = |z_1z_2|^2$ where $z_1 = a+bi$, $z_2 = c+di$. Expand z_1z_2 and read off the identity. Use this to determine which positive integers can be expressed as a sum of two squares. (*Theorem: a positive integer n is a sum of two squares iff in its prime factorisation, every prime of the form $4k+3$ appears to an even power.*)

5. Find all integers $n \geq 1$ for which n^2+n+1 divides n^3-1.

Answer: All integers $n \geq 1$.

Method: Use the identity $n^3-1 = (n-1)(n^2+n+1)$. Therefore n^2+n+1 divides n^3-1 for every integer n, hence for all $n \geq 1$.

> **Investigate further: Polynomial divisibility.** The factorisation $n^3-1 = (n-1)(n^2+n+1)$ is a special case of the cyclotomic polynomial $\Phi_3(n) = n^2+n+1$. Investigate: for which n does $\Phi_3(n) = n^2+n+1$ itself have a factor in common with n^2-1? (*Answer:* $\gcd(n^2+n+1, n^2-1) = \gcd(n+2, n^2-1)$, *which divides* 3.) Explore the analogous questions for $\Phi_4(n) = n^2+1$ and $\Phi_6(n) = n^2-n+1$.

Top 5 Patterns Today

1. **Vieta's formulas (quadratic and cubic).** $e_1 = \sum \alpha$, $e_2 = \sum \alpha\beta$, $e_3 = \alpha\beta\gamma$ read directly from the polynomial. Combine with Newton's power-sum identities to find $\alpha^k + \beta^k + \gamma^k$ without solving.

2. **Denesting nested surds.** $\sqrt{a + b\sqrt{c}} = \sqrt{p} + \sqrt{q}$ where $p + q = a$ and $4pq = b^2c$. Always try to write the radicand as a perfect square.

3. **Telescoping products and higher-order partial fractions.** $\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$. Every product of consecutive integers telescopes with the right partial fraction.

4. **Brahmagupta–Fibonacci / complex modulus.** $(a^2 + b^2)(c^2 + d^2) = (ac \pm bd)^2 + (ad \mp bc)^2$. Products of sums of two squares are themselves sums of two squares.

5. **Constraint exploitation and cyclic sum collapse.** When $x + y + z = 0$, ratios of power sums become constants. When $a + b + c = 0$: use $a^3 + b^3 + c^3 = 3abc$; when $abc = 1$: multiply fractions by abc.

Common Traps to Avoid

1. **Expanding instead of pairing.** In C2 and C5, expanding all four brackets is fatal. Always look for symmetric pairs that produce the same sub-expression.

2. **Forgetting the second Vieta formula.** $\alpha\beta + \beta\gamma + \gamma\alpha$ (not $\alpha\beta\gamma$) gives the coefficient of x in a monic cubic. Confusing e_2 and e_3 is the most common Vieta error.

3. **Taking the wrong square root in nested surds.** $\sqrt{11 - 4\sqrt{7}} = 2 - \sqrt{7}$ only if $2 - \sqrt{7} > 0$, i.e. $\sqrt{7} < 2$, which is false. The answer is $\sqrt{7} - 2$. Always check positivity.

4. **Applying AM–GM to non-positive quantities.** AM–GM ($\frac{a+b}{2} \geq \sqrt{ab}$) requires $a, b \geq 0$. Applying it when quantities can be negative gives false inequalities.

5. **Assuming $\dfrac{x}{y+z} = \dfrac{y}{x+z} = \dfrac{z}{x+y}$ forces $x = y = z$.** The case $x + y + z = 0$ (giving $k = -1$) is equally valid and must not be overlooked.

"An elegant solution is not luck — it is the reward for seeing the structure that was always there."

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other prep to look at today:
Daily Integral for Calculus and Logic